

TrES-1b

R-2305

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_110604 · created 2026-08-17T23:24:29+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455716.91707 +/- 0.00072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.116 +/- 0.018
Transit depth (Rp/Rs)^2	1.35 +/- 0.43 [%]
Orbital Inclination (inc)	89.43 +/- 0.92
Airmass coefficient 1 (a1)	0.939 +/- 0.0095
Airmass coefficient 2 (a2)	0.06 +/- 0.078
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	None
Optimal Aperture	4.45
Optimal Annulus	16.24
Transit Duration (day)	0.1009 +/- 0.0028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.116 ± 0.018 vs 0.1358 (deviation 1.1σ)
Duration fit vs published	0.1009 d vs 0.1042 d (deviation 1.17σ)
Mid-time O–C	1.8 min
Depth SNR	6.4
Residual scatter	1.02 %

Light curve

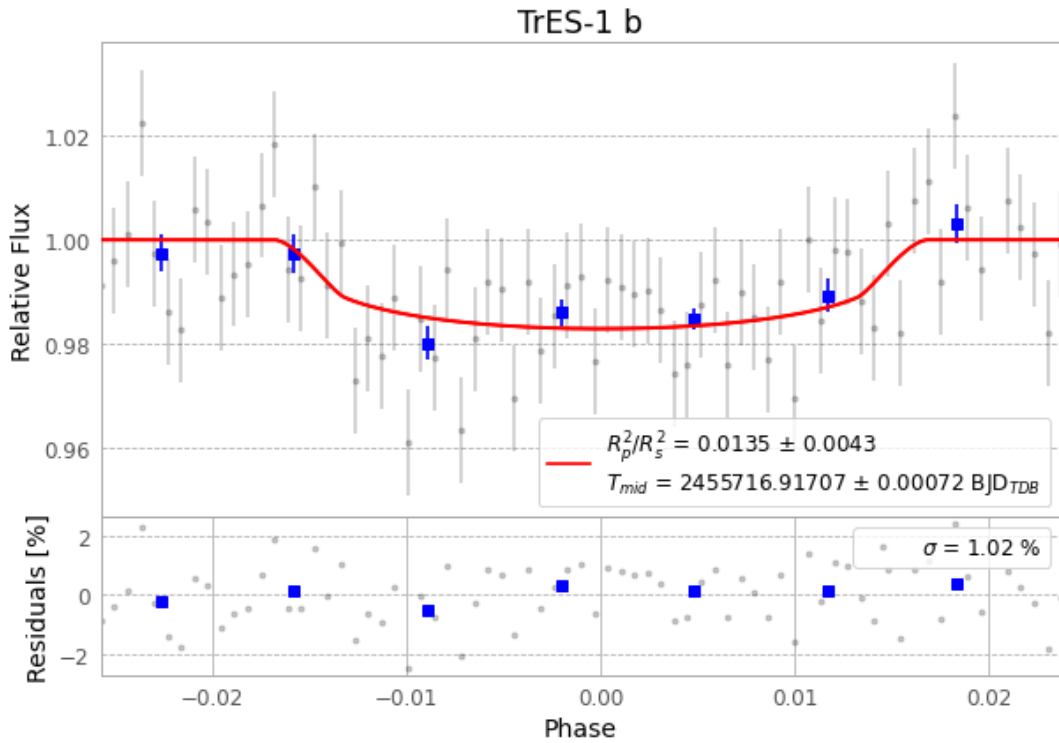


Figure 1 — FinalLightCurve_TrES-1 b_2011-06-04.png (SHA-256 7cbae7c6e80c8993...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [291, 213], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 a964b56987de99d6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

80 FITS frames (52 MB), TRES-1110604080313.FITS → TRES-1110604120013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-110604.log (1106fdbbb913...), FinalLightCurve_TrES-1 b_2011-06-04.png (7cbae7c6e80c...), FinalLightCurve_TrES-1 b_2011-06-04.csv (23f3057fe6a5...)

Compute resources

Wall time 143.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 23:24Z.